

LiSA: Light-Space Atlases for Relightable Gaussian Splatting

Supplementary Material

This supplement gives implementation details (Sec. S1), the evaluation protocol and per-scene results (Sec. S2), additional analyses (Sec. S3), registration and foreground diagnostics (Sec. S4), qualitative results (Sec. S5), and the protocols of the experiments that are still to be run (Sec. S6). Section, table and equation numbers without the prefix “S” refer to the main paper.

S1. Implementation details

Light camera. For every light we build a look-at camera from the light position $\mathbf{p}$ to the scene center $\mathbf{c}$. Its half field of view is the 99.9th percentile, over all Gaussians in front of the light, of the angular extent of their 3σ footprints, enlarged by 5% and capped at 75° ; distant floaters may therefore fall outside the atlas. Depths are measured along the optical axis, relative to $\|\mathbf{c} - \mathbf{p}\|$, and divided by the scene radius R , which is estimated from the training cameras. The texel size τ_0 is the width of one atlas texel at the depth of the scene center, in units of R , and $\tau_k = 2^k \tau_0$. The light pass uses gsplat’s classic rasterization mode (per-tile sorting by center depth, a 0.3-pixel screen-space dilation, opacities clamped at 0.99 and early ray termination), and each Gaussian contributes the depth of its center. Over 40 training lights per scene, the atlas half-angle is $8\text{--}30^\circ$ (scene medians $8\text{--}23^\circ$) and never reaches the cap; light distances are 2.0–6.2 scene radii, nearly constant within the synthetic scenes (2.8–3.5 on GS³, 2.3 on SSS-GS) and varying by up to $1.8\times$ within a real capture.

Pyramid and gather. Each level applies a separable $[1, 4, 6, 4, 1]/16$ filter with zero padding and 2×2 average pooling. Receivers sample every level bilinearly with zero padding, so points outside the light frustum see zero coverage and are treated as lit. In the descriptor, t_k is divided by four and clamped to $[-2, 2]$, and δ_k is multiplied by four and clamped to $[-4, 4]$; $M_{0,k}$ is clamped to $[0, 1]$ and the mean depth uses $\max(M_{0,k}, 10^{-4})$ in the denominator. The logit in Eq. (5) is evaluated on V_0 clamped to $[10^{-4}, 1 - 10^{-4}]$.

Networks. All networks are MLPs with SiLU activations. The flux encoder MLP ϕ has two hidden layers of width 32 and its output bias is initialized to $\text{softplus}^{-1}(1)$. The visibility residual g has two hidden layers of width 64 with a zero-initialized output layer. The transfer network h has two hidden layers of width 128 and outputs $K \cdot 3 \cdot C = 168$ values; its output bias is -7 , so that $L_{\text{tr}} \approx 0$ at the start. The material network has four hidden layers of width 128 and outputs nine RGB responses (one base and eight basis

responses); its inputs are the code $\mathbf{f}$, the normal, four-band Fourier encodings of $\mathbf{l}, \mathbf{v}, \mathbf{h}$ and the reflected view direction, five cosines $(\mathbf{n} \cdot \mathbf{l}, \mathbf{n} \cdot \mathbf{v}, \mathbf{n} \cdot \mathbf{h}, \mathbf{v} \cdot \mathbf{h}, \mathbf{l} \cdot \mathbf{v})$ and five highlight hints with $\kappa \in \{8, 32, 128, 512, 2048\}$. The eight light-independent coefficients come from a two-layer MLP of width 64 on $\mathbf{f}$ and an eight-band positional encoding, with a zero-initialized output; the lobe weights of $s(\mathbf{x})$ come from a network of the same shape with output bias -6 . The normal channels of $\mathbf{f}$ are detached wherever $\mathbf{f}$ enters a network as a latent code, so the learned normals receive gradients only through the directional terms in which they appear (cosines, the reflected view direction and the $\mathbf{n} \cdot \mathbf{l}$ input of the flux encoder).

Optimization. Densification uses the screen-space gradient rule of 3DGS (threshold $2 \cdot 10^{-4}$) from iteration 500 to 25k, with opacity resets every 3k iterations and a cap of 400k Gaussians. During the first 2k iterations, pixels outside the ground-truth mask supervise only coverage. Like all baselines, LiSA processes one training image per iteration. LDR images (real captures, SSS-GS) are linearized with $\gamma = 2.2$; clipped highlights stay clipped, and the HDR GS³ images are used as provided. The light pass is first used after the warm-up, i.e., at iteration 8001. In stage III, the networks start from their stage-II weights with fresh optimizer state; geometry, cameras and the light normalization are frozen.

Real captures. For each training camera we optimize a rotation about its calibrated optical center, starting at iteration 2k with a learning rate decayed from 10^{-3} to 10^{-5} . With fixed centers and free rotations, a global translation of the scene can be absorbed by re-aiming all cameras. After every camera update we project the rotation corrections off this three-dimensional gauge in closed form and translate the scene by the corresponding offset, which leaves the image position of the object center unchanged in every training view and keeps the reconstruction in the calibrated frame. Test cameras and lights always use the official calibration.

S2. Evaluation protocol and per-scene results

Images are evaluated at 512 px on the longest side (256 px for SSS-GS). The GS³ scenes are linear HDR renderings displayed with $\gamma = 2.2$ on a white background; the other benchmarks are displayed as provided on black. SSIM uses an 11×11 Gaussian window ($\sigma = 1.5$) with zero padding, and LPIPS uses the VGG network with inputs in $[-1, 1]$. Each method’s renderings are compared with one common set of 8-bit targets, namely the test images as encoded by our

091 evaluation pipeline. The targets produced by the baselines’
 092 own pipelines differ from these by at most one intensity
 093 level, except on three real captures (CatSmall, CupFabric,
 094 Pikachu), where 6–10% of the channel values of GS³, SSD-
 095 GS and RNG differ by up to 42 levels. Re-scoring against
 096 the common targets changed per-scene PSNR by at most
 097 0.04 dB and scene-averaged PSNR by at most 0.005 dB,
 098 and it changed no ranking. Table S1 lists all per-scene re-
 099 sults. Across scenes, the advantage of LiSA is significant
 100 against every baseline under a Wilcoxon signed-rank test
 101 ($p = 0.004, 0.007, 0.016$ and $< 10^{-4}$ against SSD-GS, GS³,
 102 OLAT-GS and RNG); on the 11 synthetic scenes it is signifi-
 103 cant against SSD-GS ($p = 0.014$) and GS³ ($p = 0.024$) but
 104 not against OLAT-GS ($p = 0.12$).

105 **Baseline runs.** All baselines use their released code and
 106 default schedules on the official training images, composite
 107 their training targets with the provided foreground masks,
 108 and render the test views at the official calibration. Four
 109 synthetic SSD-GS models (AnisoMetal, Translucent, Bunny,
 110 Dragon) come from earlier runs with the same protocol, with-
 111 out pose refinement, and were re-scored. Several RNG runs
 112 and the OLAT-GS runs on Drums and Lego reach low scores;
 113 we have not verified that these reproduce the published be-
 114 havior of the methods, which experiment E6 addresses.

115 **Use of the benchmarks during development.** The
 116 method was developed on these benchmarks. Earlier ver-
 117 sions (Tab. S3) were evaluated on the official test views, first
 118 of Cat, Pixiu, AnisoMetal, Translucent, Bunny and Dragon
 119 and later of all 18 scenes, and these results identified the
 120 failures on Lego and Drums. Design choices for the final
 121 version were then made on views held out from the training
 122 sets: the material network on held-out light groups, and the
 123 staged schedule on every tenth training view of Lego, Drums,
 124 FurBall, Cat and Soap. The final configuration was run once
 125 on all 18 scenes, with no per-scene settings and no check-
 126 point selection on test views. The results should therefore be
 127 read as benchmark results of a method developed on these
 128 benchmarks rather than as a blind test.

129 S3. Additional analyses

130 **Transfer share.** Table S2 reports, for every scene, the
 131 share of linear radiance that the final model routes through
 132 the transfer term. It is lowest on the opaque synthetic ob-
 133 jects (5.6% on Drums to 9.3% on Lego), higher on fur
 134 and the Translucent scene (about 17%), and highest on the
 135 subsurface-scattering scenes (20.5–58.0%). On real cap-
 136 tures it varies between 7.2% (FurScene) and 42.1% (Pixiu, a
 137 translucent resin figure); there the transfer term also absorbs
 138 effects that the local model cannot explain. These shares
 139 describe how the learned model allocates radiance, not a

physical decomposition.

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Light novelty of the benchmarks. The last column of
 Tab. S2 measures, for every official test view, the angle
 between its light direction and the closest training light di-
 rection, both seen from the scene origin. The median is 1.1°
 on the GS³ scenes, 0.9 – 2.3° on the real captures and 0° on
 SSS-GS, where most test lights coincide with training lights
 and only the camera–light combination is new. The bench-
 marks therefore measure relighting under densely sampled
 lights rather than extrapolation, which motivates experiment
 E1.

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Development history. Table S3 lists earlier versions of
 LiSA evaluated with the same protocol. The initial version
 used the display-domain loss and pixel receivers throughout;
 the revised version added a stronger mask loss, the highlight
 lobes and gated background supervision. The final, staged
 version is far better on the GS³ scenes but slightly behind
 the revised version on real and SSS scenes; because the two
 versions also differ in initialization, normals and receivers,
 this gap cannot be attributed to the loss domain alone.

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Training time. Table S4 lists the wall-clock training time
 of every run in the benchmark. For LiSA it covers both
 training processes including initialization and saving; for
 GS³ and SSD-GS it covers the training function up to the
 last progress record; for RNG both stages; for OLAT-GS
 both stages including mesh export, with up to 15 s of polling
 delay. GPU load and concurrency were not controlled uni-
 formly across these runs, so they only indicate the order of
 magnitude; Tab. S5 gives controlled measurements.

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S4. Registration and foreground diagnostics

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These diagnostics re-evaluate the saved test renderings of all
 five methods; no model is retrained or fitted.

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Registration (E2(b)). For every test view of the real cap-
 tures we search the integer 2D translation of the rendering
 within ± 24 px that maximizes PSNR against the common
 target, evaluated on the image with a 24 px border removed
 so that all shifts and methods use the same pixels. The base-
 lines’ renderings are best aligned by shifts of 4.0–4.6 px on
 average, against 1.7 px for LiSA (Tab. S7). After alignment,
 the mean lead of LiSA over the best baseline on the real
 captures drops from 2.0 to 0.7 dB: it remains ahead on Cat
 (+2.7 dB), Pixiu (+1.5 dB) and FurScene (+0.6 dB) and
 falls behind on CupFabric (−2.0 dB), CatSmall (−0.4 dB),
 Fish (−0.3 dB) and Pikachu (−0.1 dB). Registration thus
 explains about two thirds of the fixed-calibration margin. A

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Table S1. **Per-scene results** (PSNR $\uparrow$ / SSIM $\uparrow$ / LPIPS $\downarrow$), common targets, original test calibration, seed 0. Best PSNR per scene in **bold**.

Benchmark	Scene	RNG	OLAT-GS	GS ³	SSD-GS	LiSA (ours)
NRHints (real)	Cat	20.99 / 0.773 / 0.287	22.43 / 0.768 / 0.206	19.12 / 0.717 / 0.232	18.09 / 0.705 / 0.247	25.33 / 0.821 / 0.171
	CatSmall	18.74 / 0.792 / 0.212	31.58 / 0.949 / 0.081	33.28 / 0.967 / 0.076	33.83 / 0.967 / 0.079	33.02 / 0.961 / 0.082
	CupFabric	21.41 / 0.847 / 0.201	36.40 / 0.979 / 0.049	35.16 / 0.976 / 0.053	35.98 / 0.979 / 0.054	35.31 / 0.976 / 0.059
	Fish	24.11 / 0.842 / 0.187	23.85 / 0.834 / 0.153	22.41 / 0.809 / 0.171	23.27 / 0.822 / 0.151	28.97 / 0.896 / 0.111
	FurScene	24.12 / 0.861 / 0.162	24.77 / 0.868 / 0.113	26.09 / 0.891 / 0.103	25.15 / 0.877 / 0.112	24.47 / 0.864 / 0.119
	Pikachu	17.27 / 0.833 / 0.201	31.06 / 0.962 / 0.055	31.16 / 0.967 / 0.057	30.69 / 0.965 / 0.058	31.65 / 0.967 / 0.056
	Pixiu	19.95 / 0.848 / 0.172	20.70 / 0.841 / 0.143	23.72 / 0.864 / 0.117	23.39 / 0.862 / 0.119	26.20 / 0.881 / 0.107
GS ³ (synthetic)	AnisoMetal	24.32 / 0.923 / 0.072	27.51 / 0.961 / 0.039	27.51 / 0.950 / 0.058	28.00 / 0.957 / 0.047	27.15 / 0.956 / 0.040
	Drums	15.44 / 0.832 / 0.238	20.64 / 0.922 / 0.096	31.52 / 0.976 / 0.029	31.54 / 0.976 / 0.029	30.45 / 0.976 / 0.030
	FurBall	17.61 / 0.890 / 0.169	35.56 / 0.968 / 0.062	35.83 / 0.968 / 0.079	35.00 / 0.965 / 0.086	36.11 / 0.971 / 0.054
	Hotdog	29.40 / 0.953 / 0.060	29.67 / 0.966 / 0.046	33.32 / 0.974 / 0.036	32.94 / 0.973 / 0.037	35.22 / 0.980 / 0.029
	Lego	16.78 / 0.809 / 0.200	20.27 / 0.852 / 0.192	30.91 / 0.956 / 0.050	29.72 / 0.944 / 0.061	30.80 / 0.959 / 0.044
	Translucent	29.29 / 0.961 / 0.052	29.03 / 0.971 / 0.041	32.76 / 0.975 / 0.042	31.99 / 0.975 / 0.038	34.67 / 0.981 / 0.031
SSS-GS (synthetic)	Bunny	37.85 / 0.988 / 0.020	40.43 / 0.991 / 0.016	34.42 / 0.980 / 0.032	38.49 / 0.989 / 0.019	39.86 / 0.990 / 0.016
	Candle	41.88 / 0.994 / 0.015	45.08 / 0.997 / 0.008	38.74 / 0.988 / 0.022	38.74 / 0.989 / 0.020	45.20 / 0.996 / 0.008
	Dragon	35.23 / 0.973 / 0.042	37.79 / 0.982 / 0.026	36.10 / 0.977 / 0.032	37.38 / 0.982 / 0.027	37.38 / 0.981 / 0.026
	Soap	40.28 / 0.991 / 0.013	41.42 / 0.992 / 0.011	38.35 / 0.984 / 0.027	38.36 / 0.988 / 0.022	40.91 / 0.992 / 0.012
	Statue	30.23 / 0.968 / 0.048	38.99 / 0.989 / 0.022	36.43 / 0.985 / 0.025	35.99 / 0.983 / 0.027	39.82 / 0.989 / 0.020

Table S2. **Transfer share and light novelty per scene.** Share of linear radiance routed through L_{tr} by the final model (mean and range over 8 evenly spaced test views), mean learned visibility over covered pixels, and the angle between each official test light and the nearest training light (median / maximum).

Benchmark	Scene	Transfer share	Range	Mean V	Test-light novelty
NRHints (real)	Cat	34.2%	28–44%	0.66	2.3° / 10.5°
	CatSmall	26.8%	21–40%	0.55	1.2° / 3.5°
	CupFabric	19.2%	12–25%	0.76	1.3° / 3.6°
	Fish	28.2%	1–45%	0.88	2.2° / 8.8°
	FurScene	7.2%	5–11%	0.59	2.1° / 6.6°
	Pikachu	12.3%	9–20%	0.55	0.9° / 3.7°
	Pixiu	42.1%	37–45%	0.83	2.1° / 5.1°
GS ³ (synthetic)	AnisoMetal	7.4%	6–11%	0.89	1.1° / 3.9°
	Drums	5.6%	4–7%	0.77	1.1° / 3.9°
	FurBall	16.9%	16–19%	0.84	1.1° / 3.9°
	Hotdog	6.7%	6–8%	0.92	1.1° / 3.9°
	Lego	9.3%	7–14%	0.75	1.1° / 3.9°
	Translucent	16.7%	14–21%	0.89	1.1° / 3.9°
SSS-GS (synthetic)	Bunny	28.1%	13–81%	0.67	0.0° / 11.3°
	Candle	47.6%	28–88%	0.49	0.0° / 11.2°
	Dragon	48.2%	34–79%	0.46	0.0° / 11.2°
	Soap	58.0%	32–98%	0.63	0.0° / 11.3°
	Statue	20.5%	10–57%	0.36	0.0° / 11.0°

Table S3. **Development history** of LiSA on the same 18-scene protocol (PSNR / LPIPS). Initial: display-domain loss and pixel receivers throughout. Revised: adds a stronger mask loss, the highlight lobes and gated background supervision. Final: the staged schedule of Sec. 4, which also changes initialization, normals and receivers.

Version	Real (7)	GS ³ (6)	SSS (5)	All (18)
Initial (30k)	28.47 / 0.099	28.35 / 0.062	39.07 / 0.020	31.38 / 0.065
Revised (30k)	29.49 / 0.096	28.64 / 0.059	41.03 / 0.015	32.41 / 0.062
Final (38k)	29.28 / 0.100	32.40 / 0.038	40.63 / 0.016	33.47 / 0.056

Table S4. **Wall-clock training time** (minutes) of every run in our benchmark. Runs were not timed under strictly uniform conditions (see text); “–” marks SSD-GS runs reused without a timing log. Table S5 gives exclusive-GPU measurements.

Scene	LiSA	GS ³	SSD-GS	OLAT-GS	RNG
Cat	25.1	53.9	69.2	67.8	115.6
CatSmall	20.4	49.1	56.4	59.1	182.3
CupFabric	19.8	52.6	57.8	69.1	170.2
Fish	19.2	61.3	67.5	53.8	82.7
FurScene	20.0	54.5	60.8	70.1	81.8
Pikachu	20.5	53.0	61.7	74.8	175.9
Pixiu	19.7	50.3	59.6	55.8	80.2
AnisoMetal	28.3	63.2	–	78.3	204.5
Drums	26.2	74.8	136.6	81.6	99.6
FurBall	21.5	41.7	59.8	59.1	89.5
Hotdog	21.4	42.6	78.2	75.8	97.3
Lego	26.6	59.0	161.7	103.9	115.7
Translucent	24.1	54.7	–	71.6	237.5
Bunny	16.9	105.2	–	67.8	40.4
Candle	16.8	41.0	27.6	58.1	36.9
Dragon	16.8	43.3	–	57.8	38.2
Soap	16.6	40.6	27.4	61.8	38.5
Statue	16.7	41.3	28.8	71.3	41.1
Mean	20.9	54.5	68.1	68.8	107.1

Table S5. **Cost** on three profiled scenes, each run alone on one RTX 6000 Ada: training time in minutes (whole process, both LiSA stages) and rendering frame rate with a new light in every frame, including the light pass of LiSA.

Method	Drums		FurScene		Soap	
	min	FPS	min	FPS	min	FPS
GS ³	73.5	61	51.0	191	39.4	228
SSD-GS	141.1	29	61.2	136	26.8	198
LiSA	25.7	63	19.9	74	16.7	104

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translation is only a proxy for a pose error, so the rotation-only test-pose alignment and the retraining of E2 remain necessary.

Foreground-masked PSNR. Because LiSA supervises the rendered opacity with the foreground masks, full-image metrics might favor it through cleaner backgrounds. Table S8

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Table S6. **Appearance refinement** (stage III, 8k iterations on the same frozen 30k geometry, seed 0), averaged over Lego, Drums and Soap. Our default refines pixels in the display domain.

Receivers	Loss domain	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Gaussians	linear	34.04	.971	.036
Gaussians	display	34.24	.974	.033
pixels	linear	33.77	.973	.032
pixels	display	<u>34.05</u>	.976	.029

Table S7. **Registration diagnostic on the real captures** (E2(b)). Mean PSNR over the seven scenes with a 24 px border removed, before and after translating each test rendering by the best integer 2D shift within ± 24 px, and the mean shift length. Fixed-calibration full-image PSNR from Tab. 1 for reference.

Method	Full image	Cropped	Aligned	Shift (px)
LiSA	29.28	28.52	31.47	1.7
SSD-GS	27.20	26.45	30.76	4.0
GS ³	27.28	26.52	30.78	4.3
OLAT-GS	27.26	26.52	30.28	4.6
RNG	20.94	20.18	21.95	12.9

Table S8. **Foreground-masked PSNR** (pixels with ground-truth opacity above 0.5), scene-averaged per benchmark, against the common targets.

Method	Real (7)	GS ³ (6)	SSS (5)	All (18)
LiSA	23.67	28.23	32.76	27.72
SSD-GS	21.92	27.39	29.95	25.97
GS ³	21.98	27.83	28.91	25.86
OLAT-GS	21.97	23.30	33.01	25.48
RNG	15.58	18.25	29.42	20.31

therefore reports PSNR over the pixels whose ground-truth opacity exceeds 0.5. The ranking within each benchmark is unchanged: LiSA leads on the real captures (by 1.7 dB) and on the GS³ scenes (by 0.4 dB), and OLAT-GS leads on SSS-GS (by 0.25 dB), so the gains of LiSA do not come from cleaner backgrounds.

S5. Additional qualitative results

Figure S1 shows the atlas and the radiance components for an opaque scene with strong cast shadows and for a real translucent object, and Fig. S4 further comparisons. Figure S2 illustrates the transfer ablation of Tab. 2, and Fig. S3 the failure cases discussed in Sec. 5.6.

S6. Experiments to be run

The following experiments are referenced in the main paper with red placeholders. They use the frozen final configuration; nothing is tuned on their results.

E1: held-out light directions. Partition every training set into 30° elevation–azimuth bins of the light direction and hold out whole bins, in a fixed order, until about 10% of the training images are held out (the deterministic split implemented in our data loader). On the scenes we checked, this yields held-out lights 4–11° (median) and up to 16° away from the remaining training lights, compared with 0–2.3° for the official test splits. Train LiSA and its local-residual variant (three seeds each) and GS³, SSD-GS and OLAT-GS on the remaining images of the ablation panel (Drums, AnisoMetal, Translucent, FurScene, Soap; optionally all 18 scenes), and evaluate on the held-out images with the main protocol. Report Tab. 4, per-scene results, paired seed differences, and per-view error against angular and camera novelty; also split by light distance, include the $\mathcal{G} \equiv 0$ variant, and add a harder split that holds out a contiguous cap of lights at least 20° from all training lights.

E2: calibration-disentangled real-scene evaluation. (a) Train LiSA without the rotation refinement of training cameras on the seven real captures, and the baselines without pose refinement or with our gauge constraint, reporting the recovered pose offsets. (b) For all five methods, report secondary scores after a rotation-only test-pose alignment with the scene frozen (identical for all methods) and after the best global 2D translation within 24 px; Sec. S4 gives the latter. Fixed-calibration numbers remain the primary metric.

E3: visibility. Keep the receivers, schedule and atlas transfer fixed and replace the atlas shadow test by (a) per-Gaussian visibility splatted as an attribute, as in GS³ and SSD-GS (an option of our renderer), and (b) a hard depth shadow map with percentage-closer filtering evaluated at the pixel receivers. Also replace the Gaussian-CDF test by a Chebyshev (variance shadow map) test and sweep β , and run the single-seed rows of Tab. 2 with three seeds. Use the ablation panel plus Hotdog, Lego and one real capture, three seeds, and report paired differences with confidence intervals; for (a), also composite per-Gaussian visibility in stage III, and for (b), use an RNG-style per-pixel depth test to separate per-pixel evaluation from moment filtering.

E4: optimization ablation at the full protocol. On the ablation panel with three seeds: (a) display-domain loss in all stages; (b) no warm-up, i.e., 30k joint iterations from the hull initialization; (c) pixel receivers in stage II; (d) no stage III, evaluating the 30k checkpoint with Gaussian receivers; (e) a display-domain loss with an offset inside Γ or with RawNeRF-style reweighting [38]; (f) the staged schedule applied to GS³ or SSD-GS on LDR data.

E5: additional baselines. SSS-GS on the SSS-GS scenes once a stable reproduction under our protocol is avail-

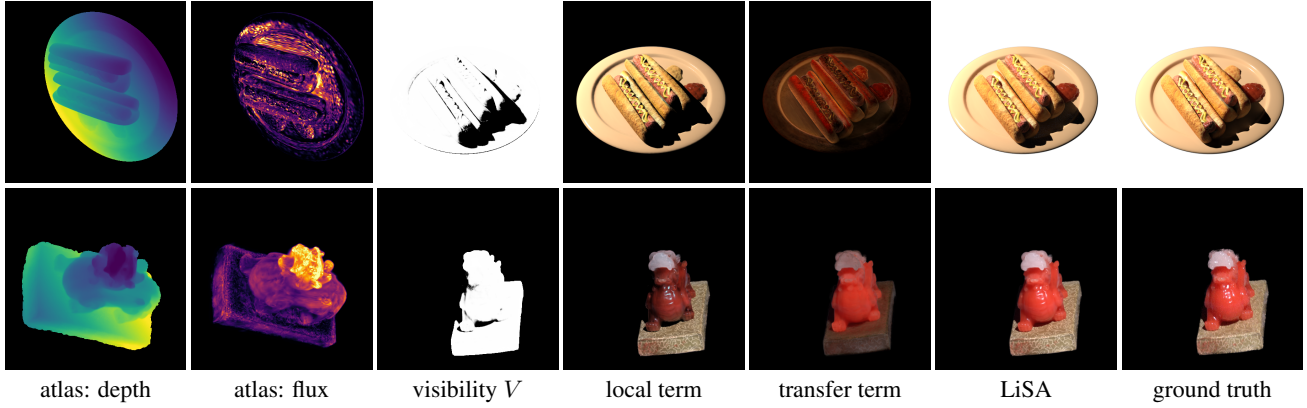


Figure S1. **Atlas and radiance components** of the final models on test views of Hotdog (top, 6.7% of the radiance through the transfer term on average) and of the real Pixiu capture (bottom, a translucent resin figure, 42.1%). On Pixiu the transfer term carries the glow of the resin, while the cast shadow on the box is explained by the visibility.

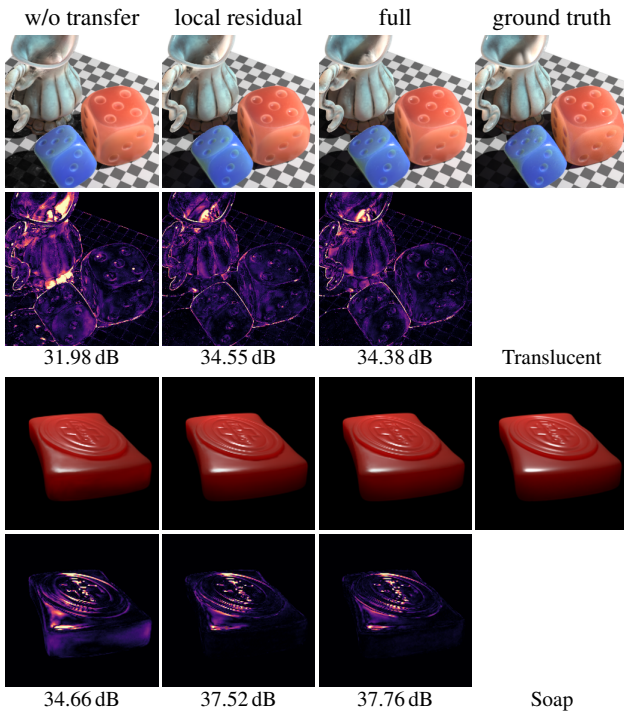


Figure S2. **Transfer ablation** on the first test view of Translucent and Soap (seed 0; PSNR of the full view; error maps show the mean absolute RGB error, saturating at 0.15). Overall brightness is nearly identical across variants, but without the transfer term the errors concentrate on the translucent bodies: the blue die, the base of the vase and the sides of the soap. The local residual and the full model have similar errors, consistent with Tab. 3.

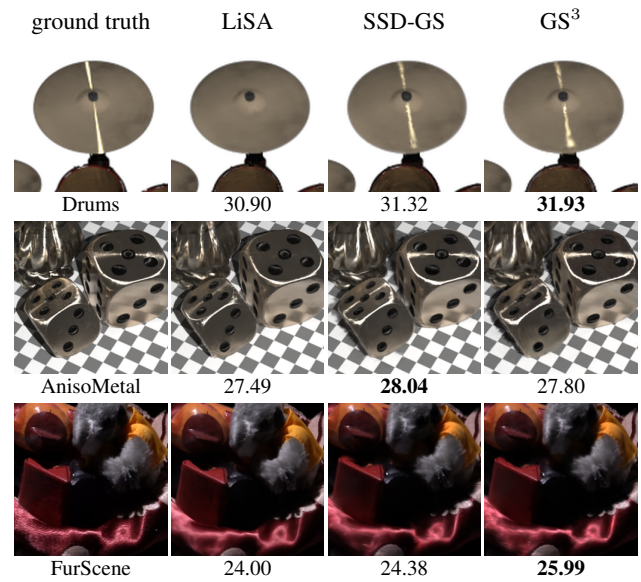


Figure S3. **Failure cases** at the median of the per-view margin (PSNR of the full view). LiSA misses the thin anisotropic streak on the cymbal, renders the brushed-metal dice too uniformly, and loses detail in real fur.

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able (the two scenes that converged can be reported now), NRHints on its real captures, and BiGS if an all-lights-on image for its initialization can be synthesized from the training data.

E6: baseline reproduction check. For each baseline, reproduce one configuration reported in its paper with the authors' protocol (resolution, image subset, background), compare with the published numbers, and inspect the low-scoring runs of RNG and OLAT-GS.

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E7: seed variance of the main comparison. Three seeds for LiSA on all 18 scenes, and for the strongest baseline at least on the ablation panel, to replace the single-run entries of Tab. 1 by means with standard deviations.

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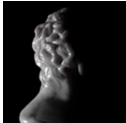
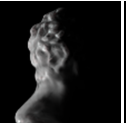
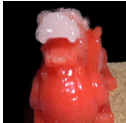
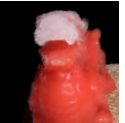
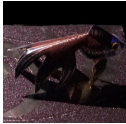
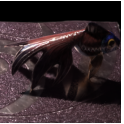
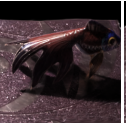
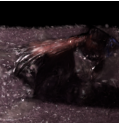
ground truth	LiSA	SSD-GS	GS ³	OLAT-GS
				
Statue	36.23	30.90	33.51	34.16
				
Pixiu	26.10	23.33	23.54	21.46
				
Fish	28.18	23.38	22.26	23.40

Figure S4. **Further comparisons** (crops; PSNR of the full view). Statue at the 75th percentile of the per-view margin, Pixiu and Fish (real) at the median. On Pixiu and Fish the baseline crops are similarly sharp but shifted relative to the target, so much of the PSNR gap reflects registration (E2).